

ATHARV KULKARNI

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[Google Scholar](#) · [LinkedIn](#) · [GitHub](#)

PUBLICATIONS

- **LLM-Symbolic Integration for Robust Temporal Tabular Reasoning.** Atharv Kulkarni, Kushagra Dixit, Vivek Srikumar, Dan Roth, Vivek Gupta. The 63rd Annual Meeting of the Association for Computational Linguistics. ACL 2025 (Findings). | [Code](#) | [Publication](#) | [Website](#) |
- **Reinforcing Code Generation: Improving Text-to-SQL with Execution-Based Learning.** Atharv Kulkarni, Vivek Srikumar. | [Publication](#) |
- **Debug or Regenerate? Test-Guided Selection for LLM Code Repair.** Atharv Kulkarni, Archiki Prasad, Vivek Srikumar, Elias Stengel-Eskin, Hyunji Lee, Mohit Bansal. The 2026 Conference on Empirical Methods in Natural Language Processing. EMNLP 2026 (Findings). | [Publication](#) | [Code](#) |
- **Agent Forest: Controlled Complexity Growth for RL environments via a multi-agent orchestration system.** Atharv Kulkarni, Santosh Borse, Hajar Emami Gohari, Mohit Bansal, Yousaf Shah. Under review at ICLR 2027

WORK EXPERIENCE

RESEARCH ENGINEER INTERN (ADVISOR: [DR. YOUSAF SHAH](#))

IBM RESEARCH – T.J. WATSON RESEARCH CENTER – GRANITE MODEL TRAINING TEAM



MAY 2026 – AUGUST 2026
YORKTOWN HEIGHTS, NY

- Developed reinforcement-learning environments and agentic datasets for training and evaluating large language models on long-horizon tool use, state tracking, reasoning, and multi-step decision making.
- First-authored “**Agent Forest: Controlled Complexity Growth for RL environments via a multi-agent orchestration system**” (under review at ICLR 2027).
- In this work we developed **Agent Forest**, a multi-agent orchestration framework that transforms seed tasks into verified, progressively harder, and more diverse agentic tasks, introducing controlled difficulty through tool expansion, branching, memory, partial observability, and compositionality.
- Expanded TAU2-Retail, TAU2-Airline, and General-Agent into hardened task collections reaching 388.6%, 338.0%, and 439.6% of their original seed-set sizes, respectively. **Demonstrated that the generated tasks introduce meaningful agentic difficulty: Qwen 3.5 397B dropped from 75.4% → 54.7% Pass@1 on TAU2-Retail; Gemma 4 31B from 66.0% → 28.6% on TAU2-Airline and 68.8% → 25.6% on General-Agent; and GLM-5.2 from 72.0% → 53.6% on TAU2-Airline.**

GRADUATE RESEARCH ASSISTANT (ADVISOR: [PROF. MOHIT BANSAL](#)) | [PUBLICATION](#) |

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL - MURGE-LAB



MAY 2025 - PRESENT
CHAPEL HILL, NC

- Developing reinforcement learning solutions for multi-agent AI in the domain of LLM based code generation with a focus on co-adversarial and curriculum-based training at the MURGE lab.
- First authored **Debug or Regenerate? Test-Guided Selection for LLM Code Repair**. In this work we developed **PATSSEL (Program-Adaptive Test-guided Repair Strategy Selection)**, a test-guided LLM code-repair framework that adaptively routes between debugging and regeneration, improving pass@1 across LeetCode, HE+Fix, and LiveCodeBench. This paper has been accepted at **EMNLP 2026 (Findings)**.

RESEARCH PROGRAMMER (ADVISOR: [PROF. DAN ROTH](#), [PROF. VIVEK GUPTA](#)) | [BLOG POST](#) | [PUBLICATION](#)

UNIVERSITY OF PENNSYLVANIA - [COGNITIVE COMPUTATION GROUP](#)



MAY 2024 – AUGUST 2025
PHILADELPHIA, PA

- **Dataset Creation:** Developed the **TEMPTABQA-C** dataset by transforming unstructured Olympic athlete info-boxes into a structured relational database, generating **1 million synthetic question-answer pairs** to evaluate LLMs in **complex temporal tabular reasoning scenarios**.
- **LLM Evaluation:** Analyzed SOTA LLMs (GPT, Llama, Gemini, etc.) across **zero-shot, few-shot, chain-of-thought, and program-of-thought prompting strategies** on TEMPTABQA-C dataset. Identified key limitations: **Counterfactual sensitivity** leads to inconsistencies when handling hypothetical or modified data. **Scalability issues** cause performance to degrade as table size grows, making it harder for models to extract relevant information. **Over-reliance on memorization** results in a significant drop in accuracy on complex queries compared to simpler ones, as LLMs often **memorize information rather than applying true reasoning**.
- **Symbolic integration:** Developed a **symbolic intermediate representation** by training LLM's to **1. Structure unstructured tabular data into a schema and 2. Translate natural language questions to SQL before execution**. This improved performance by reducing hallucinations on **counterfactual and large context scenarios** and allowing LLMs to **use SQL as a reasoning tool** thereby increasing accuracy from **59% to 73%**.
- **Publication to ACL:** First authored “**LLM-Symbolic Integration for Robust Temporal Tabular Reasoning**”, introducing **TEMPTABQA-C**, a benchmark dataset and **symbolic approach** to advance LLMs' temporal reasoning. **This paper is accepted in ACL Findings 2025.**

NLP RESEARCH ASSISTANT (ADVISOR: [PROF. VIVEK SRIKUMAR](#)) | [PUBLICATION](#)

UNIVERSITY OF UTAH - [UTAH NATURAL LANGUAGE PROCESSING GROUP](#)



AUGUST 2024 – AUGUST 2025
SALT LAKE CITY, UT

- **Reinforcement Learning:** Developed a **neuro-symbolic RL** framework that embeds a **live SQL execution engine within the LLM training loop**, enabling models to learn **NL-to-SQL** generation directly from **execution-based scalar rewards and real database feedback**.
- **Reward Engineering:** Designed an **adaptive reward function** that integrates syntactic correctness, Exact Match Score (EMS), Relaxed Exact Match Score (REMS), chain-of-thought reasoning, and schema constraint adherence to guide model training towards precise SQL outputs.
- **Training:** Applied **Group Relative Policy Optimization (GRPO)** to optimize LLM policy updates for **generating structured SQL**, handling multi-step temporal reasoning, and adapting to diverse query complexities. Utilized the TEMPTABQA-C benchmark to systematically train and evaluate models on complex, counterfactual, and multi-step temporal queries.
- **Low Rank Adaptation & Multi-GPU Training:** Employed **parameter-efficient fine-tuning (PEFT)** via **Low-Rank Adaptation (LoRA)**, leveraging high-performance **multi-GPU infrastructure (NVLink)** to scale training efficiently while minimizing computational costs.
- **Performance Benchmarking:** **RL Training** allowed **SQLCoder-7B** to improve **Exact Match Score (EMS)** from **31.49% to 49.83%** and **reduce syntactic error rate from 25.43% to 14.71%** on the TEMPTABQA-C benchmark, **nearly matching the EMS of the much larger SQLCoder-70B (52.08%)**.
- **Publication:** First authored “**Reinforcing Code Generation: Improving Text-to-SQL with Execution-Based Learning**”, introducing the RL pipeline and the trained model.

EDUCATION

MASTER OF SCIENCE (THESIS), COMPUTER SCIENCE | (ADVISOR: [PROF. MOHIT BANSAL](#))

AUGUST 2025 – MAY 2027

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

- **Coursework:** Graduate Research, Transfer Learning, Efficient Deep Learning, Neuro-Symbolic AI Systems, Multimodal AI.

BACHELOR OF SCIENCE (THESIS), MAJOR: COMPUTER SCIENCE | MINOR: PHYSICS | (ADVISOR: [PROF. VIVEK SRIKUMAR](#))

AUGUST 2021 – MAY 2025

UNIVERSITY OF UTAH

- **Coursework:** Machine Learning, Natural Language Processing, Deep Learning, Advance AI (Reinforcement Learning), NeuroSymbolic Modelling, High Dimensional Data Analysis, Database Systems, Visualization for Data Science, Algorithms, Data Structures, OOP, Computer Systems, Computer Organization, Probability and statistics, Linear Algebra, Discrete structures, Computational Physics, Software Practice.
- **Undergraduate Research Opportunity Program (UROP) Award** – Awarded the UROP award for conducting research under Prof. Vivek Srikumar and Prof. Vivek Gupta for the publication: “**Reinforcing Code Generation: Improving Text-to-SQL with Execution-Based Learning.**”.
- **Thesis: Improving Complex Reasoning Skills In LLMs Through Tool-Based Neuro-Symbolic Methods** [\[Link to Thesis\]](#)

PROJECTS

1. **Machine Learning: Predicting Court Decisions from Historical Trial Data (Tech Stack: Python, Pandas, Matplotlib, GitHub)** | [Link to Code](#) |
 - **Developed Machine Learning Library:** Built a comprehensive machine learning library from scratch by implementing RandomForest, Logistic Regression, Perceptron, SVM and a custom Neural Network. Used TF-IDF and categorical features along with the machine learning library to predict Old Bailey court decisions.
 - **Achieved 3rd Place** on Kaggle with a score of 0.81 by leveraging the custom library to predict legal outcomes from noisy, sparse data.
2. **Natural Language Processing (Tech Stack: Python, PyTorch, Pandas, Jupyter Notebook, GitHub)** | [Link to Code](#) |
 - **Feedforward Neural Networks & Word Embeddings:** Trained a Deep Averaging Network using GloVe embeddings in PyTorch, optimized word vectors, implemented batching, and developed a misspelling correction system to enhance model accuracy and robustness.
 - **Transformer Language Modeling:** Built and extended a Transformer model in PyTorch for sequence classification and language modeling, utilizing multi-head attention, positional encodings, and causal masking to achieve high performance in sequence prediction.
3. **Reinforcement Learning (Tech Stack: Python, PyTorch, NumPy, Pandas, Matplotlib, Jupyter Notebook, OpenAI Gym, MuJoCo, GitHub)** | [Link to Code](#) |
 - **Behavioral Cloning & Behavioral Cloning from Observations (BCO)** [\[Link to Report\]](#): Built imitation-learning pipelines with an inverse-dynamics model, achieving an 83% success rate on MountainCar using five expert demonstrations.
 - **Multi-Armed Bandit Simulator** [\[Link to Report\]](#): Implemented and benchmarked ϵ -greedy and UCB1 strategies across large-scale Gaussian and Bernoulli bandit simulations.
 - **Markov Decision Process (MDP) Solver with Value & Policy Iteration** [\[Link to Report\]](#): Developed value iteration and policy iteration solvers for stochastic grid-world environments with noisy transitions and obstacles.
 - **Tabular Q-Learning & Deep Q-Networks (DQN)** [\[Link to Report\]](#): Applied tabular Q-learning to Blackjack and implemented a DQN with experience replay and target-network updates, exceeding human-level LunarLander performance.
 - **Policy Gradient Methods (Discrete and Continuous Control)** [\[Link to Report\]](#): Built REINFORCE and Actor-Critic agents for discrete and continuous control, improving CartPole returns from approximately 21 to 294 and extending to MuJoCo control tasks.
4. **Deep Learning (CS230) - Stanford University (Tech Stack: Python, NumPy, CUDA, Matplotlib, Pandas, Jupyter, HPC, Hyperparameter Optimization)** | [Link to Code](#) |
 - Developed and optimized k-Nearest Neighbors, Linear Classifiers, Support Vector Machines, and Softmax Classifiers for image classification.
 - Built and fine-tuned Fully Connected Neural Networks from scratch using Backpropagation, Batch Normalization, Dropout, and Regularization.
 - Implemented Convolutional Neural Networks (CNNs) for CIFAR-10 and ImageNet with architectures like GoogleNet and ResNet.
 - Applied Recurrent Neural Networks (RNNs), LSTMs for sequential data, and developed GANs (Style-GAN, Alias-free GAN) for synthetic data generation.
 - Utilized Autoencoders for feature extraction, Multi-Task Learning (MTL) for multi-objective performance, and Word Embeddings for NLP applications.
5. **Database Management System for Learning Management System (LMS) Software. (Tech Stack: Java, MySQL, MariaDB, Azure SQL Database, GitHub)** | [Link to Code](#) |
 - **Phase 1 - ER Modeling:** Created an ER model to define entities and relationships for the LMS, ensuring data integrity and normalization.
 - **Phase 2 - SQL Tables:** Implemented normalized tables, enforcing business rules with primary and foreign keys.
 - **Phase 3 - SQL Queries:** Developed and optimized SQL queries to efficiently retrieve data within the LMS, including stored procedures and triggers, to automate course enrollment, grade management, and assignment distribution. The queries ensured real-time data processing and seamless user interactions within the LMS.
 - **Phase 4 – Deployment:** Deployed the database on Azure SQL Database, integrating it with the LMS application, while ensuring network security and performance.
6. **Enhancing Text-to-SQL with Schema-Grounded Symbolic Intermediate Representations** | [Link to Code](#) | [Link to Preprint](#) |
 - Developed a neuro-symbolic Text-to-SQL pipeline using a Schema-Grounded Symbolic Intermediate Representation (SIR) to improve schema adherence, SQL executability, and interpretability on unseen databases.
 - Fine-tuned Qwen3-4B with LoRA and GRPO using rewards for symbolic validity, schema grounding, execution success, and answer correctness.
 - Achieved 72.82% execution accuracy on Spider Dataset, while increasing SQL schema validity to 98.45%.